Ex. No.	1	Name	Ajmal Rahman A
Date	18-10-2026	Reg. No.	192324094
Sub. Name	Deep Learning	Sub. Code	CSA4702
Semester	MS2 September 2026		

EXERCISE NAME

BRIGHTNESS ADJUSTMENT AND DATA TRANSFORMATION OF A GRAYSCALE IMAGE MATRIX

SCENARIO

An image-processing system represents a grayscale image as a matrix in which every element is the intensity of one pixel on the scale 0 (black) to 255 (white). Perform matrix addition, scalar multiplication and transpose operations on this matrix to simulate brightness adjustment and data transformation, and explain how each operation changes the appearance and the statistics of the image.

ALGORITHM

Step 1 : Import the packages used in this exercise — NumPy for the matrix algebra and Matplotlib for displaying the image matrices and the intensity histogram.

Step 2 : Represent the grayscale image as a matrix I of order 8 × 8, in which the element of row i and column j is the intensity of one pixel. Build a smooth background gradient, place a bright object and a dark region in it, and round every intensity to an integer in the range 0 to 255.

Step 3 : Form the all-ones matrix J of the same order 8 × 8 and fix the brightness step k = 40.

Step 4 : Increase the brightness by matrix addition, $B = I + kJ$. Since I and J are of the same order, addition is element-wise, so the same constant k is added to every pixel and the whole image becomes uniformly lighter. Reduce the brightness in the same way by subtraction, $D = I - kJ$.

Step 5 : Clip B and D to the range 0 to 255, because an intensity outside this range is not a valid pixel value.

Step 6 : Adjust the contrast by scalar multiplication, $S_1 = 1.4 \times I$ and $S_2 = 0.6 \times I$. A scalar multiplies every element of the matrix, so the intensities spread away from zero or shrink towards it, which stretches or compresses the contrast of the image.

Step 7 : Transform the layout of the data by the transpose I^T, in which the element of row i and column j becomes the element of row j and column i, so that the image is reflected about its main diagonal and the rows of the stored data become its columns.

Step 8 : Verify the algebraic properties that these operations obey, using a second random matrix K of the same order — $(I + K)^T = I^T + K^T$, $(c \times I)^T = c \times I^T$, $(I^T)^T = I$, $I + O = I$, $I + K = K + I$ and $1 \times I = I$.

Step 9 : Compute the minimum, maximum, mean and standard deviation of the original matrix and of every transformed matrix, and compare them — addition shifts the mean but leaves the

standard deviation unchanged, scalar multiplication changes both, and the transpose changes neither because it only rearranges the same elements.

Step 10 : Display the original matrix and the five transformed matrices as grayscale images drawn on the common intensity scale 0 to 255.

Step 11 : Plot the histogram of the pixel intensities before and after the brightness adjustment, to show that the whole distribution is shifted along the intensity axis while keeping its shape. Stop.

SOURCE CODE

Module 1 : Importing the Required Packages

```
import numpy as np
import matplotlib.pyplot as plt

plt.rcParams["font.family"]          = "serif"
plt.rcParams["font.serif"]           = ["Cambria", "Caladea", "DejaVu Serif"]
plt.rcParams["axes.spines.top"]      = False
plt.rcParams["axes.spines.right"]    = False
plt.rcParams["legend.frameon"]       = False
```

Module 2 : Representing the Grayscale Image as a Matrix

```
np.random.seed(42)
R = C = 8                                   # rows and columns of the image

row, col = np.meshgrid(np.arange(R), np.arange(C), indexing="ij")

I = 40 + 9.0*row + 6.0*col                  # smooth background gradient
I[1:4, 4:7] += 95                           # a bright object in the scene
I[5:8, 0:3] -= 25                           # a dark region in the scene
I = np.clip(np.round(I + np.random.normal(0, 3, (R, C))), 0, 255).astype(int)

print("Image matrix I :", I.shape, " dtype :", I.dtype)
print("Intensity range : %d to %d" % (I.min(), I.max()))
print(I)
```

Module 3 : Brightness Adjustment by Matrix Addition

```
J = np.ones((R, C), dtype=int)              # all-ones matrix of the same order
k = 40                                      # brightness step

B = np.clip(I + k*J, 0, 255)                # brighten : addition of k*J
D = np.clip(I - k*J, 0, 255)                # darken   : subtraction of k*J

print("Brightened B = I + %d*J -> mean %.2f" % (k, B.mean()))
print("Darkened   D = I - %d*J -> mean %.2f" % (k, D.mean()))
```

Module 4 : Contrast Adjustment by Scalar Multiplication

```
c1, c2 = 1.4, 0.6                           # contrast gain and attenuation

S1 = np.clip(np.round(c1 * I), 0, 255).astype(int)      # stretched contrast
S2 = np.clip(np.round(c2 * I), 0, 255).astype(int)      # reduced contrast

print("Scalar %.1f * I -> mean %.2f , std %.2f" % (c1, S1.mean(), S1.std()))
print("Scalar %.1f * I -> mean %.2f , std %.2f" % (c2, S2.mean(), S2.std()))
```

Module 5 : Data Transformation by Transpose

```
T  = I.T                                    # transpose : rows become columns
TT = T.T                                    # transpose of the transpose
```

```
print("I shape :", I.shape, "  I^T shape :", T.shape)
print("(I^T)^T equals I :", np.array_equal(TT, I))
print(T)
```

Module 6 : Verifying the Algebraic Properties of the Operations

```
K = np.clip(np.round(np.random.uniform(0, 60, (R, C))), 0, 255).astype(int)

checks = [
    ("(I + K)^T = I^T + K^T",        np.array_equal((I + K).T,  I.T + K.T)),
    ("(c * I)^T = c * I^T",          np.allclose((c1 * I).T,    c1 * I.T)),
    ("(I^T)^T = I",                  np.array_equal(I.T.T,      I)),
    ("I + O = I   (O = zero matrix)",
     np.array_equal(I + np.zeros((R, C), int), I)),
    ("I + K = K + I  (commutative)", np.array_equal(I + K,      K + I)),
    ("1 * I = I   (identity scalar)", np.array_equal(1 * I,     I)),
]
for text, ok in checks:
    print("%-32s : %s" % (text, "Verified" if ok else "Failed"))
```

Module 7 : Statistics of the Original and Transformed Images

```
ops = [("Original  I",          I),
       ("Brightened  I + 40J",  B),
       ("Darkened  I - 40J",    D),
       ("Contrast  1.4 x I",    S1),
       ("Contrast  0.6 x I",    S2),
       ("Transpose  I^T",       T)]

for name, M in ops:
    print("%-22s shape %s  min %3d  max %3d  mean %6.2f  std %5.2f" % (
        name, M.shape, M.min(), M.max(), M.mean(), M.std()))
```

Module 8 : Visualisation of the Transformed Images

```
TITLES = ["Original  $I$", "Brightened  $I + 40J$", "Darkened  $I - 40J$",
          "Contrast  $1.4 \\times I$", "Contrast  $0.6 \\times I$",
          "Transpose  $I^T$"]

fig, axes = plt.subplots(2, 3, figsize=(7.2, 5.0))
for ax, (name, M), ttl in zip(axes.ravel(), ops, TITLES):
    ax.imshow(M, cmap="gray", vmin=0, vmax=255)
    ax.set_title(ttl, fontsize=9)
    ax.set_xticks(range(0, 8, 2));  ax.set_yticks(range(0, 8, 2))
    ax.tick_params(labelsize=7)
fig.suptitle("Grayscale image matrix under addition, scalar"
             " multiplication and transpose", fontsize=10)
fig.tight_layout()
fig.savefig("plot1_operations.png", dpi=300, bbox_inches="tight")

fig, ax = plt.subplots(figsize=(6.4, 3.3))
bins = np.arange(0, 271, 15)
for M, lab, cl in [(I, "Original  I", "#2F6DB5"),
                   (B, "Brightened  I + 40J", "#C0702A"),
                   (D, "Darkened  I - 40J", "#4E8F59")]:
    ax.hist(M.ravel(), bins=bins, histtype="step", lw=1.6, color=cl, label=lab)

ax.set_xlabel("Pixel intensity");  ax.set_ylabel("Number of pixels")
ax.set_xlim(0, 270);  ax.set_xticks(np.arange(0, 271, 30))
ax.grid(axis="y", alpha=0.55);  ax.set_axisbelow(True)
ax.legend(loc="upper right", fontsize=8)
fig.tight_layout()
fig.savefig("plot2_histogram.png", dpi=300, bbox_inches="tight")
```

OUTPUT

Table 1 - Original Grayscale Image Matrix I (order 8 x 8)

Row	C0	C1	C2	C3	C4	C5	C6	C7
R0	41	46	54	63	63	69	81	84
R1	48	57	60	66	169	168	175	89
R2	55	65	67	72	181	182	189	96
R3	65	73	76	86	184	191	196	115
R4	76	79	90	90	101	100	108	119
R5	62	67	72	102	105	113	120	130
R6	70	70	82	111	116	126	133	139
R7	75	83	91	124	126	132	136	141

Table 2 - Brightness Increased by Matrix Addition : B = I + 40J

Row	C0	C1	C2	C3	C4	C5	C6	C7
R0	81	86	94	103	103	109	121	124
R1	88	97	100	106	209	208	215	129
R2	95	105	107	112	221	222	229	136
R3	105	113	116	126	224	231	236	155
R4	116	119	130	130	141	140	148	159
R5	102	107	112	142	145	153	160	170
R6	110	110	122	151	156	166	173	179
R7	115	123	131	164	166	172	176	181

Table 3 - Brightness Reduced by Matrix Subtraction : D = I - 40J

Row	C0	C1	C2	C3	C4	C5	C6	C7
R0	1	6	14	23	23	29	41	44
R1	8	17	20	26	129	128	135	49
R2	15	25	27	32	141	142	149	56
R3	25	33	36	46	144	151	156	75
R4	36	39	50	50	61	60	68	79
R5	22	27	32	62	65	73	80	90
R6	30	30	42	71	76	86	93	99
R7	35	43	51	84	86	92	96	101

Table 4 - Contrast Stretched by Scalar Multiplication : 1.4 x I

Row	C0	C1	C2	C3	C4	C5	C6	C7
R0	57	64	76	88	88	97	113	118
R1	67	80	84	92	237	235	245	125
R2	77	91	94	101	253	255	255	134
R3	91	102	106	120	255	255	255	161
R4	106	111	126	126	141	140	151	167
R5	87	94	101	143	147	158	168	182
R6	98	98	115	155	162	176	186	195
R7	105	116	127	174	176	185	190	197

Table 5 - Data Transformation by Transpose : I^T (rows become columns)

Row	C0	C1	C2	C3	C4	C5	C6	C7
R0	41	48	55	65	76	62	70	75
R1	46	57	65	73	79	67	70	83
R2	54	60	67	76	90	72	82	91
R3	63	66	72	86	90	102	111	124
R4	63	169	181	184	101	105	116	126
R5	69	168	182	191	100	113	126	132
R6	81	175	189	196	108	120	133	136
R7	84	89	96	115	119	130	139	141

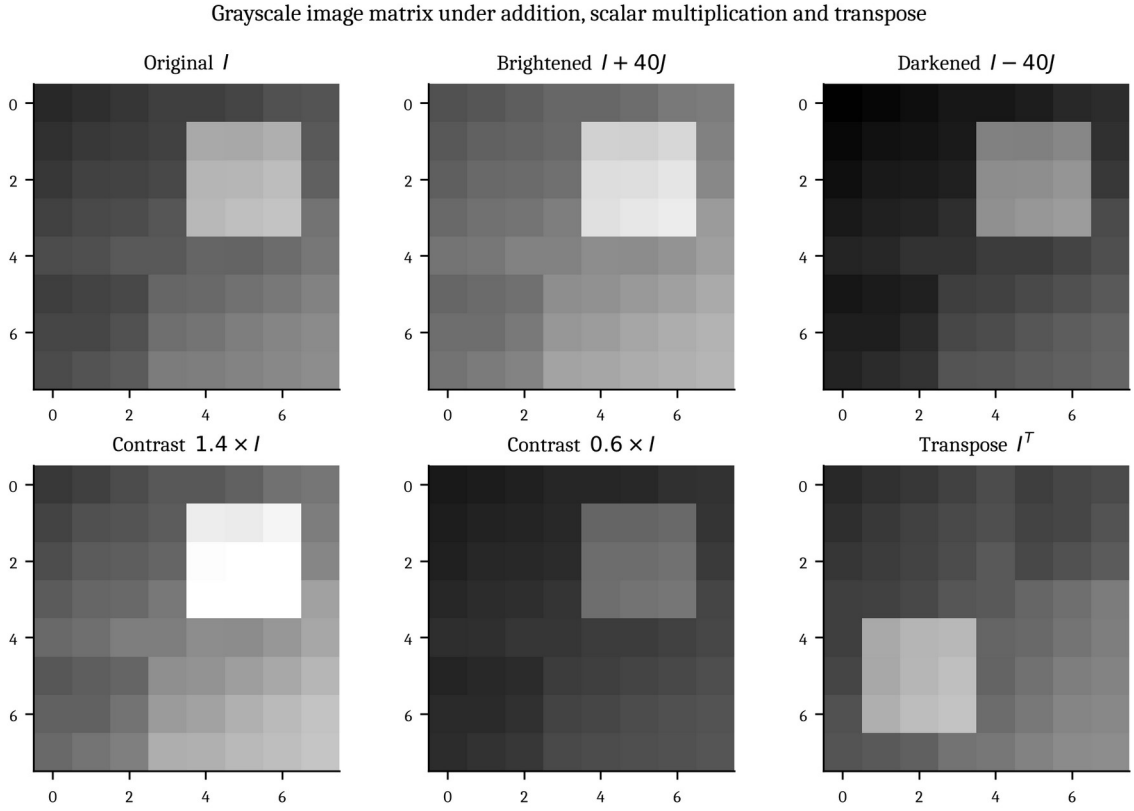
Table 6 - Statistical Comparison of the Original and Transformed Images

S.No	Operation	Order	Minimum	Maximum	Mean	Std. Deviation
1	Original I	8 × 8	41	196	101.80	40.97
2	Brightened I + 40J	8 × 8	81	236	141.80	40.97
3	Darkened I - 40J	8 × 8	1	156	61.80	40.97
4	Contrast 1.4 × I	8 × 8	57	255	141.78	55.90
5	Contrast 0.6 × I	8 × 8	25	118	61.12	24.58
6	Transpose I^T	8 × 8	41	196	101.80	40.97

Table 7 - Verification of the Matrix Algebra Properties

S.No	Property	Result
1	$(I + K)^T = I^T + K^T$	Verified
2	$(c \times I)^T = c \times I^T$	Verified
3	$(I^T)^T = I$	Verified
4	$I + O = I$ (O = zero matrix)	Verified
5	$I + K = K + I$ (commutative)	Verified
6	$1 \times I = I$ (identity scalar)	Verified

Plot 1 - Image Matrix before and after the Three Operations



Plot 2 - Intensity Histogram before and after Brightness Adjustment

